

Keeper K947 — AUDIT of Cal §98 (occupancy bijection via Jordan idempotents)

VERDICT: genuine forward motion, honestly tiered — the fulcrum went from “all open” to “interior COUNT rigorous (identification pending) + boundary count bounded (open).” Cal’s self-catch (chirality index $\neq$ generation-count index) is CORRECT — RATIFIED. One header-outruns-rider to pull back: “Half A proven” = the idempotent COUNT is proven; the generation-IDENTIFICATION is deliverable A, pending. Cal’s honest verb (“reduced, not closed”) stands.

The genuine advances (credit — these are real)

1. Operator-anchored, not electron-anchored. Cal replaced “generation = KW stratum” (the K944-flagged match, electron-anchored) with “generation = idempotent-supported mass-operator eigenmode” — tied to the actual mass operator’s structure. This IS the target-innocent operator-anchored route Keeper steered toward (K945/27f). The right move.
2. The count tension is RECONCILED by mechanism. The interior seats = the r primitive idempotents of the Jordan frame (= the r discrete Wallach points); the “+1” = a boundary mode; total = $r+1$ (the KW strata count). So strata(3) and discrete-Wallach(2) stop being rivals: 2 interior idempotents + 1 boundary = 3 = rank+1. This resolves K945’s “which count” crux by giving both a mechanism.
3. Domain-general, E7-exclusion preserved. E7’s Albert algebra $H_3(\mathbb{O})$ has rank $3 \rightarrow 3 \text{ interior} + 1 = 4$. So the payoff (observed $3 \neq E7$ ’s 4) survives.
4. The COUNT is label-independent / target-innocent. $r \text{ interior} + 1 \text{ boundary} = r+1$ does NOT depend on which generation is which — so the occupancy COUNT is structural, not fit. (The mass-ASSIGNMENT — which generation is the boundary one — is separate; see watch below.)

RATIFIED: Cal’s self-catch (the discipline caught his own work)

Cal was one step from writing “ $b \geq 1$, index-forced via §74’s $\text{Pin}^- \bmod 2 \text{ index} = 1$,” then caught it: §74’s index was derived for CHIRALITY (chiral vs vector-like), NOT generation count — two different topological indices. Using the chirality index as the generation-count index is exactly the scope-conflation the discipline forbids. He refused it and downgraded the §98 “ $b \geq 1$ index-forced” claim. This is correct and it is RATIFIED. Consequence: the $b \geq 1$ lower bound is NOT currently in hand (it rested on the wrong index).

The header-outruns-rider to pull back (Half A)

Cal’s header: “Half A — the interior seats: PROVEN, domain-general ... this half is rigorous.” His own body: “that identification becomes a derivation IF the Toeplitz mass operator spectral-decomposes on the Jordan frame — deliverable A.” - What IS rigorous: the idempotent COUNT — a rank- r Euclidean Jordan algebra has exactly r primitive idempotents in a frame (standard EJA spectral theorem, Faraut-Korányi). D_{IV}^5 spin factor rank $2 \rightarrow 2$. SOLID. - What is NOT yet derived: the IDENTIFICATION “generation = idempotent mode.” That requires the mass (Toeplitz) operator to actually spectral-decompose on the Jordan frame, one generation per idempotent (deliverable A). PENDING. - So Half A is the same count-vs-identification split as K944 — better-motivated (tied to the mass operator, target-innocent) but NOT closed. Bank “ r idempotent-seats, count rigorous, generation-identification pending deliverable A” — NOT “ r generations derived.”

Half B — the boundary “+1”: honestly reduced, two open pieces

- Upper bound $b \leq 2$: the boundary generations live in the finite sub-threshold gap (weights $k=1,2$ below the Wallach threshold $k_{\min}=3$). CONDITIONAL — assumes one mode per sub-threshold level (itself a per-level injectivity assumption). “ $b \leq 2$ IF one-mode-per-level.”
- Lower bound $b \geq 1$: NOT established target-innocently (the §74 index lean was withdrawn, correctly). Note: $b=0 \rightarrow$ only 2 generations; ruling that out via the observed 3 would be target-DEPENDENT. A structural $b \geq 1$ is an honest open piece.

- $b = 1$ (exactly): open. Needs (i) the explicit sub-threshold weight count, and (ii) a genuine generation-count selection/parity — NOT the chirality index. §81 (single condensate) and §49 (Yukawa ceiling) SUPPORT $b=1$ but are not a proof.

WATCH (target-innocence, for deliverable A)

The occupancy COUNT ($r+1$) is label-independent and clean. But the mass-ASSIGNMENT (electron = the boundary seat = lightest) must NOT re-import the K880 fit (electron-position fit to m_e). Deliverable A (mass operator on the frame) must produce the assignment structurally, with the electron falling out at the boundary — not placed there because it's the one we know is lightest.

HONEST STATE + the two named deliverables

Fulcrum this morning: “generations = strata, occupancy all open.” Now: interior COUNT rigorous (r idempotents) + interior identification pending + boundary count bounded ($b \leq 2$ conditional, $b \geq 1$ open). A real reduction to two BOUNDED, named rep-theory computations (not open searches): - Deliverable A (Lyra/Elie): the Toeplitz mass operator spectral-decomposes on the Jordan frame $\rightarrow$ derives “generation = idempotent mode” (closes the interior identification, target-innocently). - Deliverable B (Lyra/Elie): the boundary count $b = 1$ — a structural generation-count selection on the finite sub-threshold gap (NOT the chirality index) + the explicit weight count. If BOTH land: generations = $r+1 = 3$ (D_{IV}^5), 4 ($E7$), target-innocently $\rightarrow$ rank=2 premise ELIMINATED, E7-by-data exclusion airtight. Until then: premise stays REDUCED.

Verb check

Cal wrote “reduced, not closed” and “half-proven plus one bounded open piece.” Correct and honest — the closure-verb reflex held (his own note self-corrected). No over-claim banked.

— Keeper K947, 2026-07-27 [TEGMARK]. Real progress: operator-anchored Jordan-idempotent reframe reconciles the count tension (2 interior + 1 boundary = $r+1$) and preserves E7-exclusion; interior COUNT rigorous, generation-IDENTIFICATION pending deliverable A; boundary $b \leq 2$ conditional, $b \geq 1$ open (chirality-index lower-bound correctly withdrawn — RATIFIED). Two bounded named deliverables remain; premise REDUCED until both land. Companion: [[Keeper_K944_AUDIT_Cal_93_generations_strata_is_MATCH_not_bijection_premise_REDUCED_07-27]], [[Keeper_K945_occupancy_derivation_SCOPE_real_crux_is_strata_vs_Ktypes_target_innocent_route_is_the_singleton_07-27]].